LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

MNM TEAM

Scientific Work

From research question to seminar paper

Prof. Dr. Dieter Kranzlmüller PD Dr. Andre Luckow

MNM-Team, LMU München

Seminar Emerging Topics in ML and AI, November 3, 2026

Agenda

1

Foundations of Science

Empirical and applied research, hypotheses, the scientific method and AI in research

2

The Seminar Paper

From topic to research question, paper structure and the golden thread

3

Literature

Finding, accessing and reading scientific papers

4

Writing Well

Writing advice, including AI writing

Scientific Work

Credits and Sources

Special thanks go to Prof. Kranzlmüller, Dr. Schiffers and Sophia Grundner-Culemann for permission to use material from the MNM course on scientific work.

SOURCES AND REFERENCES

- Kranzlmüller, Schiffers, Wissenschaftliches Arbeiten und Lehren, LMU, 2017
- Werner Sesink, Einführung in das wissenschaftliche Arbeiten, 2012
- Martin Böttcher, Einführung in das wissenschaftliche Arbeiten, Universität Leipzig
- Grundlagen Wissenschaftliches Arbeiten, Universität Paderborn
- Christina Huber, Gregor Imhof, Nicole Müller, Einführung in das wissenschaftliche Arbeiten
- Bettina Schnor, Thomas Ludwig, Vortragstechnik, Universität Potsdam (backup slides)

Foundations of Science

Empirical and Non-Empirical Science

Empirical science

- Phenomena are investigated through **experiments**, **observation** or **questioning**
- Knowledge must hold up against data

Non-empirical science

- Knowledge is gained **without direct observation**
- Knowledge must hold up against logic and proof

Computer science combines both: complexity theory and formal verification are non-empirical; benchmarks, training runs and user studies are empirical.

Source: Kranzlmüller and Schiffers, Wissenschaftliches Arbeiten (WS 2015/16).

Foundations of Science

Basic Research and Applied Science

Basic research

Scientific definition, review and discussion of principles

principles, theories, methods

Applied science

Concrete problem solving in technology and business using the scientific method

problems from practice

new knowledge flows back

Practice

The starting point of applied science is close to practice

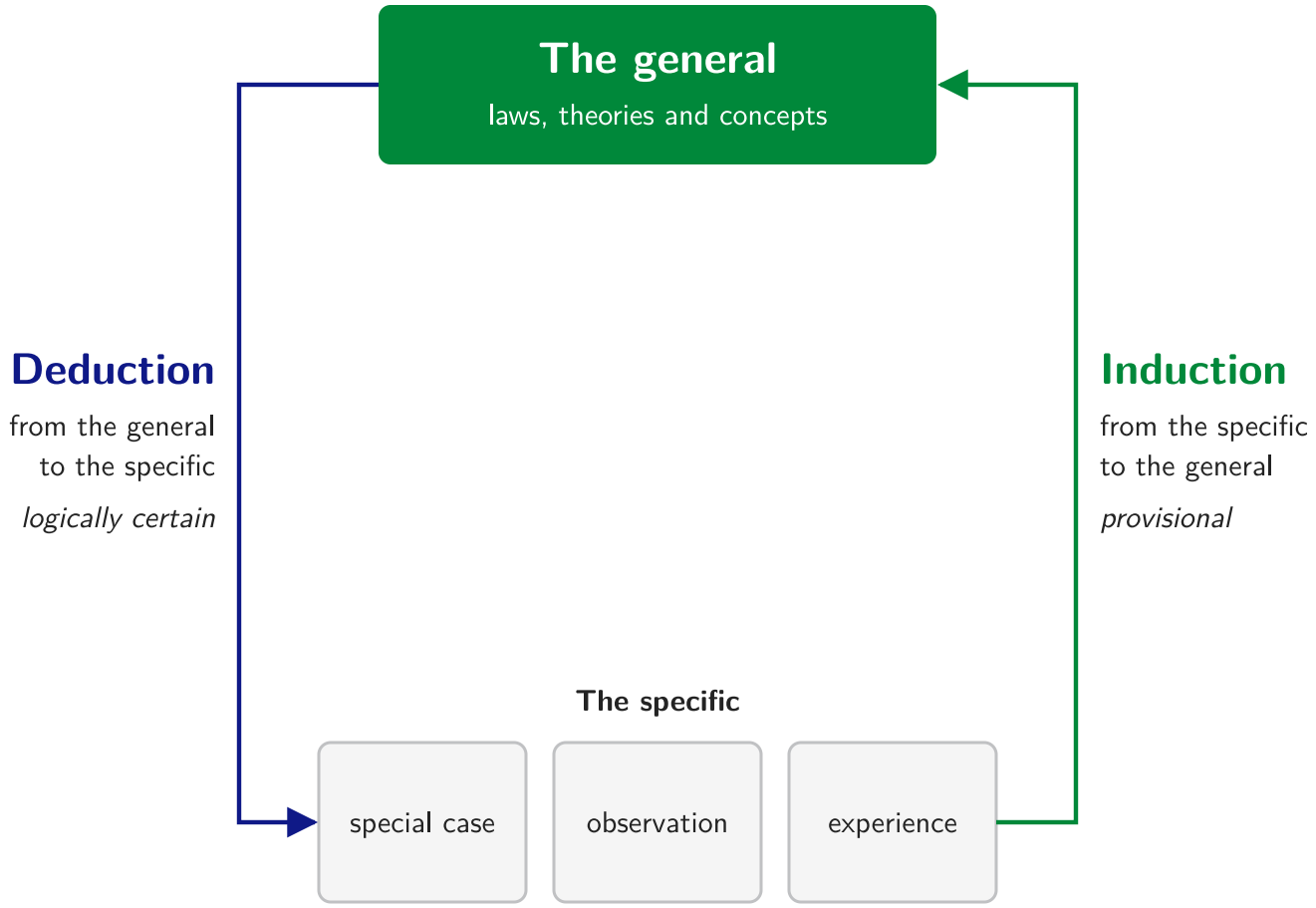
EXAMPLE FROM MACHINE LEARNING

Basic: how does model error scale with parameters, data and compute?

Applied: which model size gives the best quality for our fixed GPU budget?

Source: Kranzlmüller and Schiffers, Wissenschaftliches Arbeiten (WS 2015/16).

Foundations of Science
Deduction and Induction



- **Deduction:** derives special cases from general laws; if the premises are true, the conclusion is true
- **Induction:** generalizes from observations and experiences to general concepts; conclusions stay probable and provisional
- **Analysis** (breaking a subject into parts) and **synthesis** (combining parts into a whole) can use either form of reasoning

> Typically, both methods are used **complementarily**: always check one against the other.

Source: Kranzlmüller and Schiffers, Wissenschaftliches Arbeiten (WS 2015/16).

Foundations of Science

From Hypothesis to Theory to Law

STEP 1

Hypothesis

A logical but **unproven** explanation for a given set of facts; the starting point for further experiments and observations.

Hypotheses must be **testable**: testing means comparing observations with the predictions of the hypothesis.

STEP 2

Theory

A hypothesis that has been **tested multiple times**, explains previous observations and accurately predicts future observations.

STEP 3

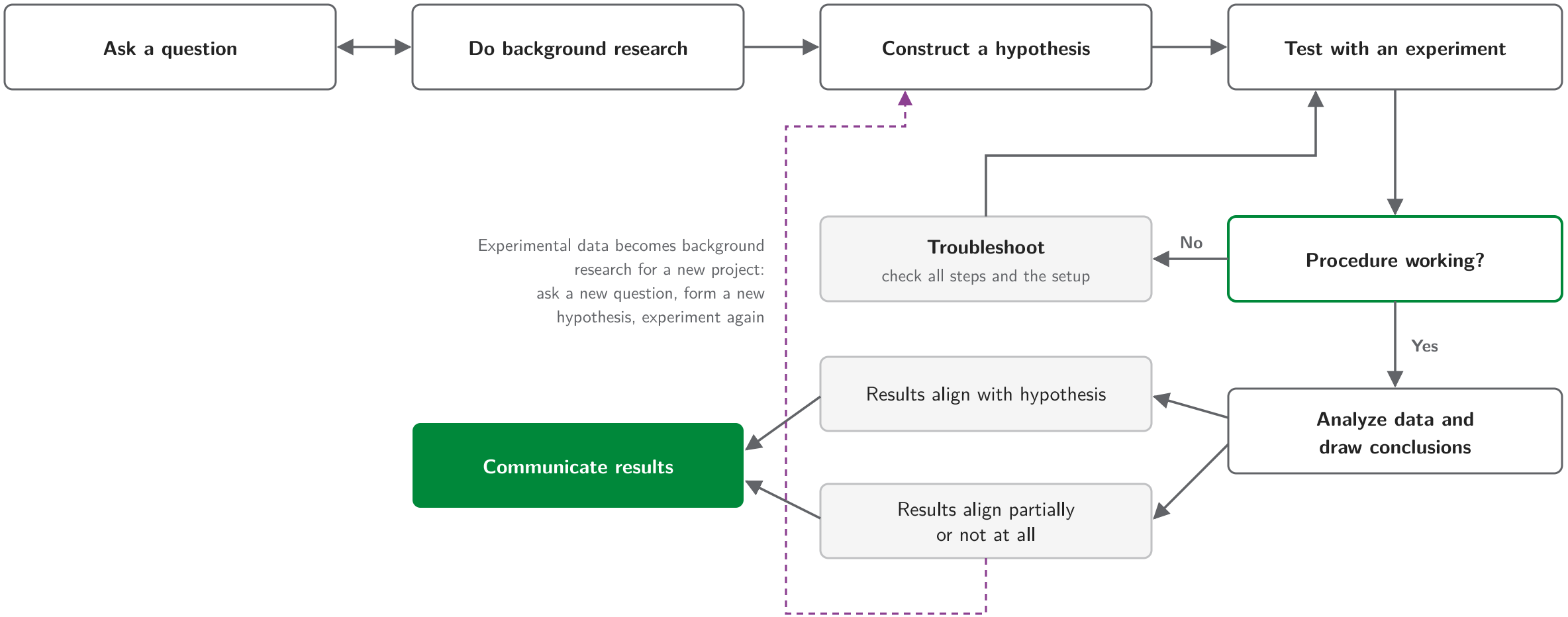
Law

A theory that has been **extensively tested and never falsified**, explains previous observations and accurately predicts future observations.

Mind the difference between the everyday word **"theory"** (an unproven guess) and the **scientific term** (a well-tested explanation).

Source: Kranzlmüller and Schiffers, Wissenschaftliches Arbeiten (WS 2015/16).

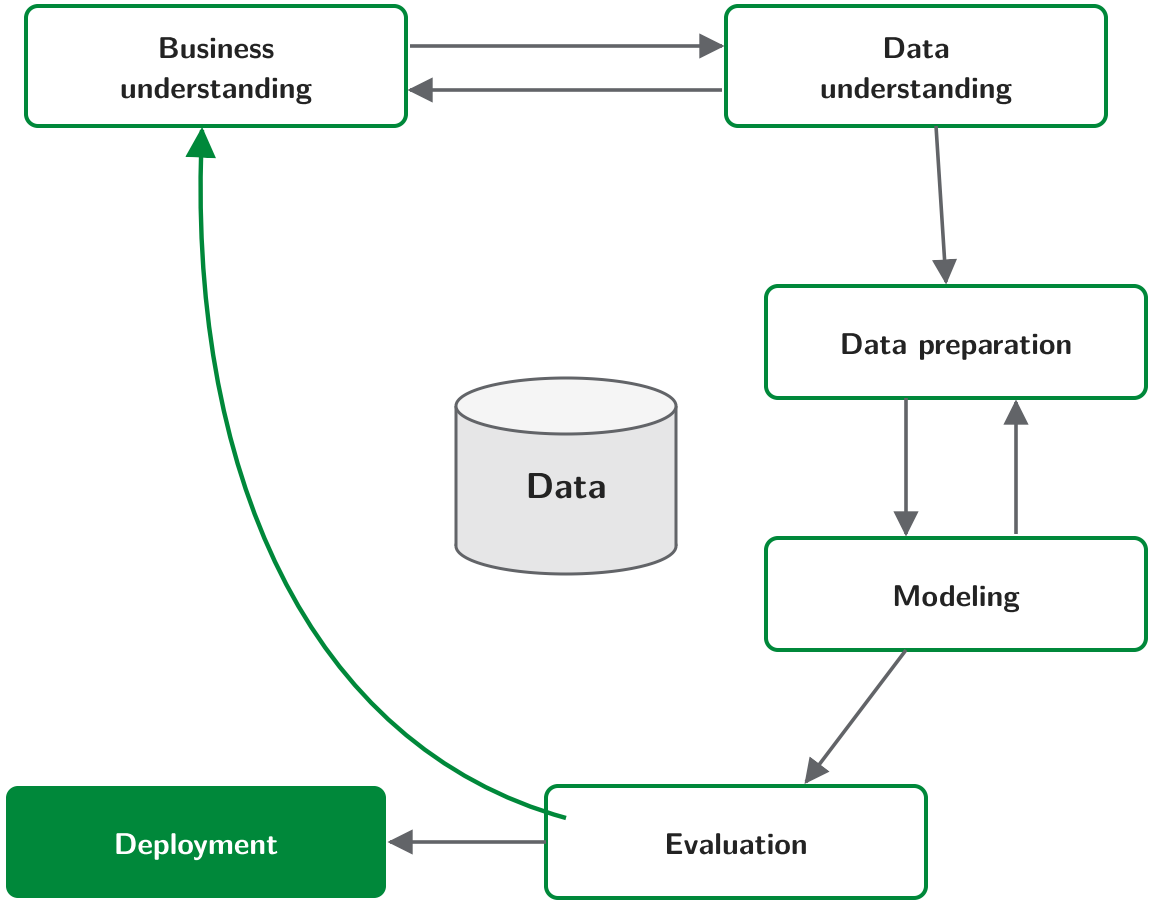
Foundations of Science
The Scientific Method



Redrawn after Science Buddies, Steps of the Scientific Method.

Foundations of Science

Data Science Process: The CRISP-DM Model



CROSS-INDUSTRY STANDARD PROCESS FOR DATA MINING

- Six phases around the data, from understanding the problem to deployment
- **Iterative:** arrows lead back, e.g., from modeling to data preparation
- **Evaluation decides:** deploy the model or return to business understanding

> Applied science in practice: many ML papers report their experiments along these phases.

Source: Provost and Fawcett (2013), Data Science for Business, O'Reilly; Chapman et al., CRISP-DM 1.0.

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Foundations of Science

Science in the Age of AI: Three Examples from 2026

MNM TEAM

OPENAI, MAY AND AUGUST 2026

Erdős unit-distance conjecture

- A general-purpose reasoning model **disproved** a conjecture from 1946
- External mathematicians checked the proof and wrote a human-readable version
- In August: ten more results on open problems

OPENAI, SEPTEMBER 2026

Navier–Stokes blow-up

- About 10,000 agents worked for 88 hours: a 166-page proof plus a Lean formalization
- Covers only flows with an **external force**; the unforced problem remains open
- **Contested:** does the Lean code match the paper, and where did the idea come from?

GOOGLE, NATURE, JUNE 2026

Empirical Research Assistance (ERA)

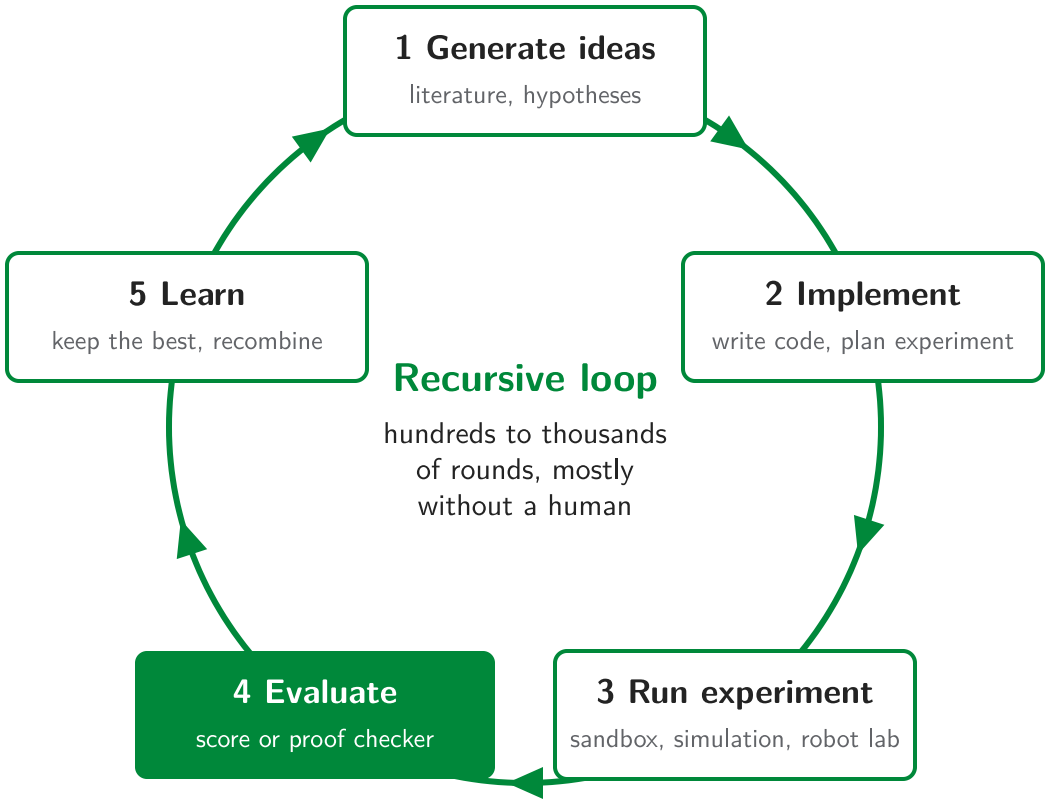
- An LLM with tree search rewrites code to **maximize a benchmark score**
- 40 single-cell methods beat the public leaderboard; 14 COVID-19 forecasts beat the CDC ensemble
- The authors' caveat: optimizing a score is not yet scientific discovery

AI advances fastest where a result can be **checked automatically**, by a proof checker or a benchmark score. "What really matters is the size of the gap between proposing an answer and finding out whether it's any good." (Adam Brown, quoted in Socher 2026)

OpenAI, unit-distance disproof (May 2026), ten advances (Aug. 2026), Navier–Stokes (Sep. 2026); Aygün et al., An AI system to help scientists write expert-level empirical software, Nature 654 (2026); R. Socher, The Eureka Machine, PublicAffairs (2026).

Foundations of Science

Closing the Loop: Automated Research Cycles



- **Empirical Research Assistance (ERA)** runs this loop for scientific software: the LLM rewrites code, a sandbox scores it, tree search picks the next candidate. Exploring a set of ideas takes **hours instead of months**.
- **Eureka Machine** (Socher): a knowledge model, a physics model, simulations and robot labs; lab results flow back and update the models.
- **Recursive:** AlphaEvolve and the Darwin–Gödel Machine also improve the code that does the improving.

> Humans still set the **question** and the **score**. The loop delivers answers; Terence Tao warns of a "decoupling between getting answers and getting **understanding**".

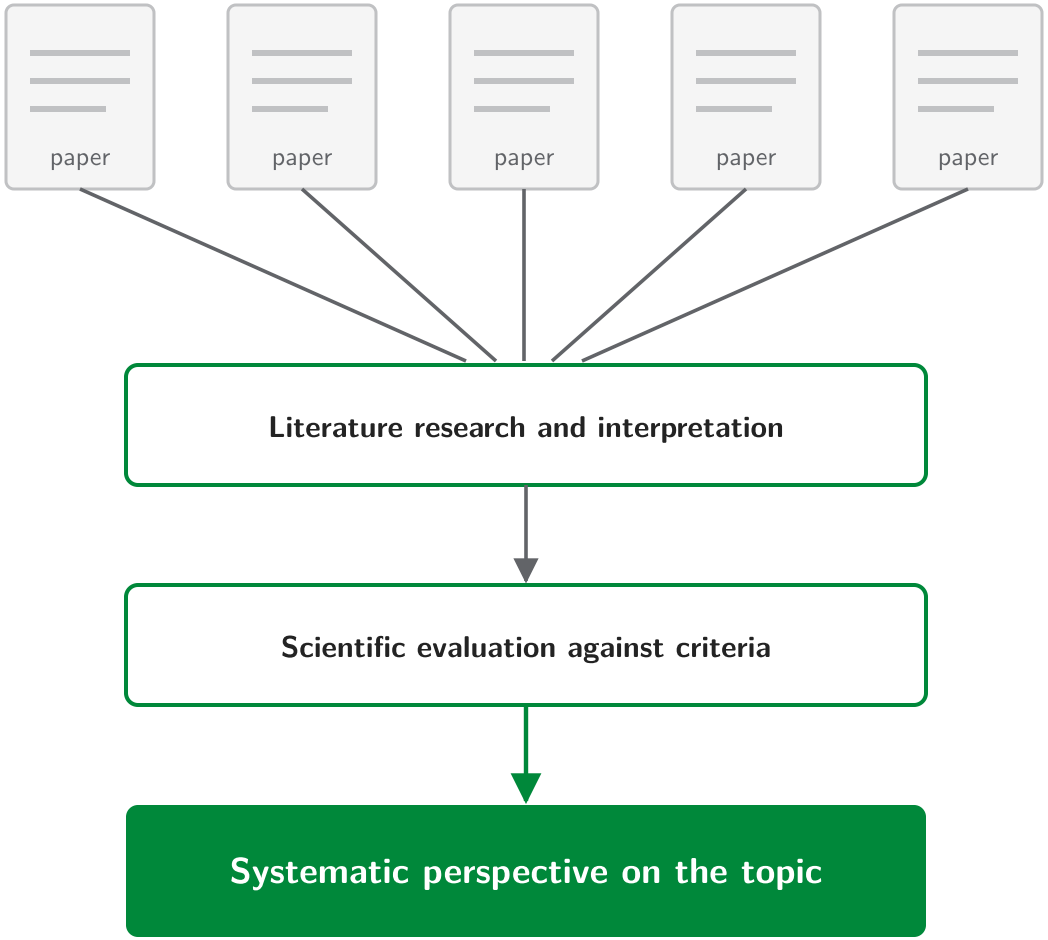
Aygün et al., Nature 654 (2026), Fig. 1; R. Socher, The Eureka Machine (2026), ch. 8; T. Tao in New Scientist (Sep. 2026).

The Seminar Paper

Aim of a Seminar Paper

1. Apply the concepts of scientific work and write a **research paper**
2. Focus on **literature research**, its interpretation and its scientific evaluation
3. Develop a **systematic perspective** on a topic

> A seminar paper is **more than a summary** of the core papers.



The Seminar Paper

From Topic to Research Question

START

Broad topic

The given topic definition is deliberately very general

STEP 1

Initial literature research

The core papers are only a starting point

STEP 2

Clarify and focus

Narrow the scope; ask your mentor for feedback

GOAL

Research question

Define a research question or hypothesis

EXAMPLE: TOPIC 14, LLM ROUTING AND MODEL SELECTION

"LLM routing" as a broad topic → *How much inference cost can learned routers save compared with always calling the strongest model, and at what loss in answer quality?*

The Seminar Paper

Structure of a Seminar Paper

1. Introduction

- **Motivation:** why should someone read the paper? Why is the topic relevant?
- Which questions or hypotheses does the paper cover?
- Background information on the topic
- Classification in the overall context
- Why is the topic relevant for this audience (students, MNM-Team)?
- Structure of the paper

2. Methodology

- Methods used to investigate the topic
- Criteria used for the comparison (as table or figure)

3. Description of the topic

- Overview and current state of research
- Based on primary literature
- Structured, not a random list of facts

4. Evaluation and discussion

- Critical reflection and evaluation against the defined criteria
- Identified gaps in the results
- Recommendations for further action

5. Conclusion and future work

- What does it all mean?
- Which questions remain open?

Adapted from J. Leek, firstpaper; NeuroData, paper template; TU Dresden, Writing a term paper.

The Seminar Paper
Citation

1. **Cite all your sources.** Anything else is plagiarism.
2. Find the **primary sources** for important theoretical foundations and technologies.
3. Use the given **bibliography style** (IEEE).

IEEE STYLE IN THE TEXT

The Transformer architecture **[1]** replaced recurrence with self-attention.

IEEE STYLE IN THE BIBLIOGRAPHY

[1] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 5998–6008.

Check every reference yourself. AI tools produce plausible-looking citations to papers that do not exist.

The Seminar Paper

Your Paper Needs a Golden Thread

1	2	3	4	5	6	7	8	9
Motivation	**Background**	**Related work**	**Demarcation**	**Question**	**Method**	**Results**	**Conclusion**	**Future work**
Why care?	What readers need to know	What others did	What is in scope	What you ask	How you answer	What you found	What it means	What remains open

Start with the **motivation** and the **"why"**!

RUNNING EXAMPLE FOR THE NEXT SLIDES

A toy research problem that everyone understands: *How can eggs be replaced to make vegan chocolate chip oatmeal cookies?*

Source: S. Grundner-Culemann, Grading criteria for seminars, MNM-Team.

The Seminar Paper

1. Motivation: Why Should Anyone Care?

- What is the **greater context** of the problem?
- Why do we want to find an answer? Pure interest, benefit from a solution, avoiding harm, ...

RUNNING EXAMPLE

Why replace eggs at all? Environmental impact, animal welfare or health reasons.



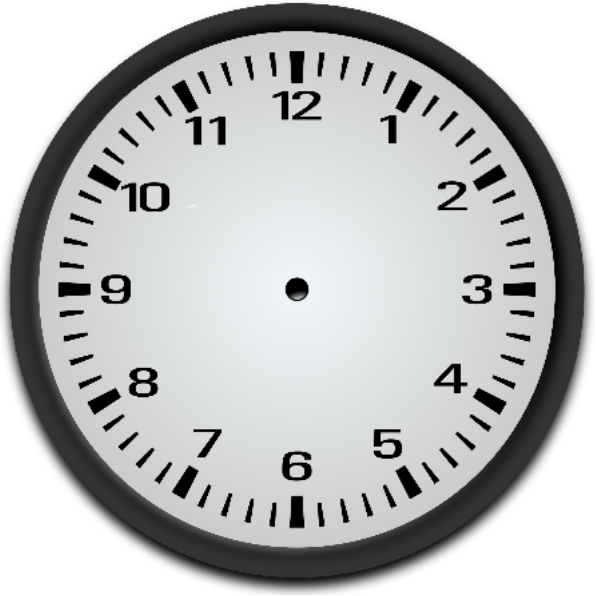
Same apple, different context: gravity or harvest?

The Seminar Paper

2. Background: What Does the Reader Need to Know?

- Explain the background for the **intended audience**
- Your audience: your **peers** in the seminar and the MNM-Team
- Which field are they from? What do they need to know to follow your argument?

> **EXAMPLE**
> A paper on post-quantum cryptography does not need to explain everything about qubits.



"Tell me the time,
don't build me the clock."

The Seminar Paper

3. Related Work and Research Gap

- Which other solutions to this or similar questions exist?
- Why are they insufficient, or which role does your own work play in this context?

RUNNING EXAMPLE

Eggs have been replaced in muffins, cakes, pancakes, tortillas and protein shakes. In cookies, known substitutes make the dough very crumbly.

						2		
3	9		2					
			6		7			4
		9				5		
4								8
		7				3	4	
5			4	9				
	7			5	3		2	
		3						

☐ Filled digits: what others have already solved

The Seminar Paper

4. Demarcation of Your Own Work

- Which aspects will you address, and which not?
- How does the topic relate to the conference theme?

RUNNING EXAMPLE

In scope: chocolate chip oatmeal cookies.
Out of scope: rye cookies and "healthy" low-sugar variants.

						2		
3	9		2					
			6		7			4
		9				5		
4								8
		7				3	4	
5			4	9				
	7			5	3		2	
		3						

Green: the part of the problem you work on

The Seminar Paper

5. Scientific Question

- State the question **clearly**
- Divide it into **sub-questions**

> A good question can be answered with your method and within **6 to 8 pages**.

How can eggs be replaced to make vegan chocolate chip oatmeal cookies?

Sub-question 1
Which functions do eggs fulfil in cookie dough (binding, moisture)?

Sub-question 2
Which plant-based substitutes provide these functions?

Sub-question 3
How do the substitutes compare in texture and taste?

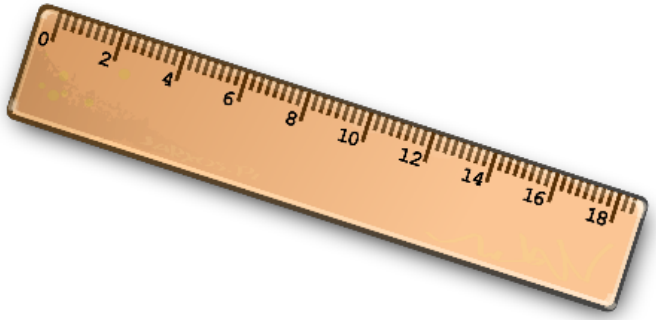
The Seminar Paper
6. Method

- With which **scientific approach** is the problem solved?
- Why is this approach **suitable**?

RUNNING EXAMPLE

Bake test batches with each substitute under identical conditions; measure spread and breaking strength; run a blind taste test.

In a seminar paper, the method is usually a **structured literature analysis**: search strategy, selection criteria and comparison criteria.



The Seminar Paper

7. Results and Observations

- What results did the method yield?
- What is the answer to the scientific question?
- Further discussion and analysis

> Make the evaluation **traceable**: readers must be able to follow how you got from the data to the answer.

7	**4**	**6**				2		
3	9	**5**	2					
2	**1**	**8**	6		7			4
		9				5		
4								8
		7				3	4	
5			4	9				
	7			5	3		2	
		3						

Bold green digits: your results

The Seminar Paper

8. Conclusion and 9. Future Work

Conclusion

- What does it all mean?
- **Distill** the findings, do not only summarize them

Future work

- Which aspects remain unanswered?
- Which new questions do the results prompt?
- Do not stop thinking at the solution

RUNNING EXAMPLE

Does the best substitute also work for rye cookies?

7	**4**	**6**				2		
3	9	**5**	2					
2	**1**	**8**	6		7			4
		9				5		
4								8
		7				3	4	
5			4	9				
	7			5	3		2	
		3						

your answer the next open question

The Seminar Paper
On Novelty



Received: 31 October 2017 | Revised: 16 March 2018 | Accepted: 25 March 2018

DOI: 10.1002/fsn3.656

ORIGINAL RESEARCH

WILEY Food Science & Nutrition Open Access

Effect of total replacement of egg by soymilk and lecithin on physical properties of batter and cake

Sara Hedayati | Mostafa Mazaheri Tehrani

Department of Food Science and Technology, Ferdowsi University of Mashhad (FUM), Mashhad, Iran

Correspondence
Sara Hedayati, Department of Food Science and Technology, Ferdowsi University of Mashhad (FUM), Mashhad, Iran.
Emails: sara.hedayati@mail.um.ac.ir; sara_hedayatiii@yahoo.com

Abstract

The baking industry is interested in finding cost-effective and healthful alternatives for eggs. Therefore, in this study the effects of total replacement of egg by soymilk (SM) in combination with 0–6% soy lecithin (SL) on batter (density, microstructure, viscosity, and texture) and cakes (height, volume, density, texture, color parameters, and sensory attributes) were determined and compared with cakes manufactured with eggs. The results showed that all batters had shear thinning behavior and pro-



Trends in Food Science & Technology 111 (2021) 346–359

Contents lists available at ScienceDirect

Trends in Food Science & Technology

journal homepage: www.elsevier.com/locate/tifs

ELSEVIER

A review of egg replacement in cake production: Effects on batter and cake properties

Gamze Nil Yazici *, Mehmet Sertac Ozer

Department of Food Engineering, Faculty of Agriculture, Cukurova University, Adana, Turkey

ARTICLE INFO

Keywords:
Egg replacement
Eggless cake

ABSTRACT

In recent years, there is a growing demand for and interest in egg replacement, due mainly to health problems (phenylketonuria, egg allergy) or health-related concerns (cholesterol, avian influenza), dietary preferences (vegan, vegetarian), and/or religious beliefs of consumers; and also to issues concerning sustainable food supply

Hedayati and Mazaheri Tehrani (2018), Food Science & Nutrition; Yazici and Ozer (2021), Trends in Food Science & Technology.

- Is your research going to be "new"?
- How do you come up with a new question? When is your question clear at all?

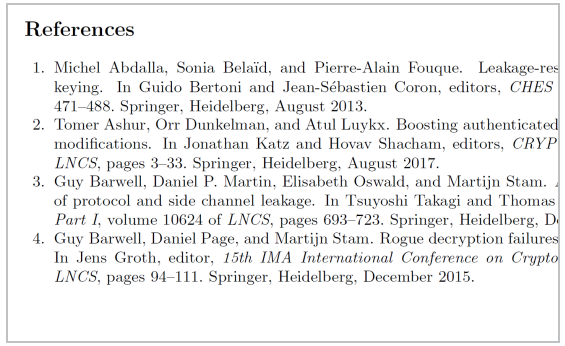
> Egg replacement in **cakes** is well studied. Novelty can lie in the gap: a new setting, a new criterion or a **systematic comparison**.

Literature

How to Find (Good) Literature



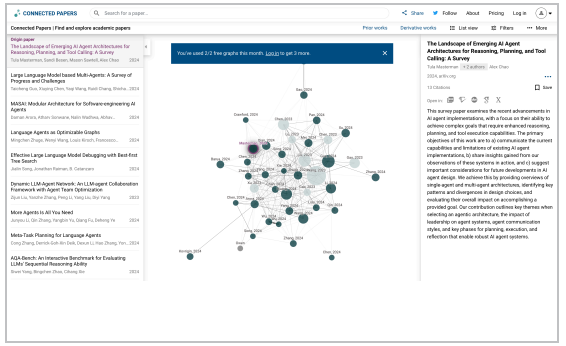
Google Scholar: keyword search; "Cited by" leads forward to newer work



Reference lists: follow the citations of a good paper backward (snowballing)



arXiv: the newest preprints; check whether a paper was later peer-reviewed

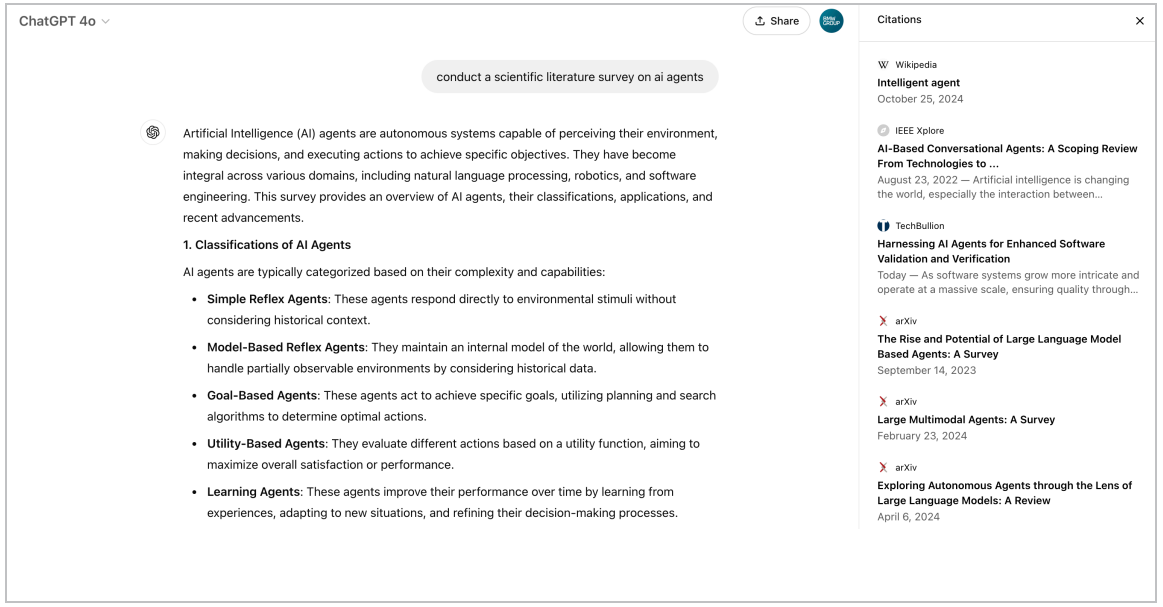


Connected Papers: explore the citation graph around a seed paper

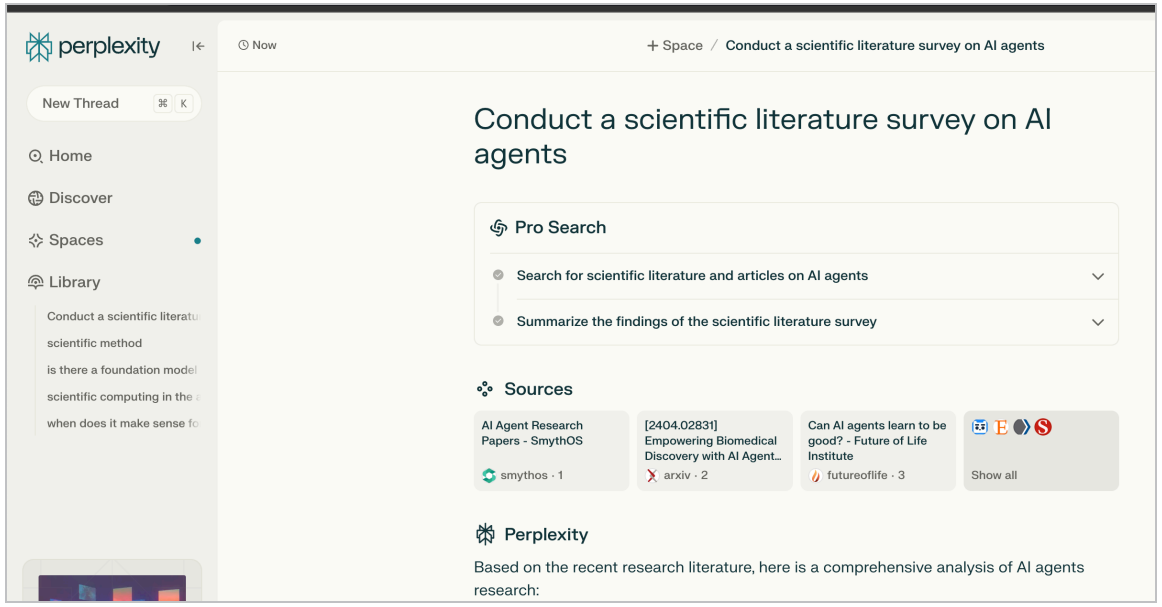


> Use a general search engine for **orientation** (e.g., "best conferences in AI") and Google Scholar for the **scientific question** itself (e.g., "does deep learning solve AI").

Literature
AI Research Assistants



ChatGPT with web search



Perplexity Pro Search

Prompt in both tools: "Conduct a scientific literature survey on AI agents."

- Useful for **orientation** and first pointers to papers

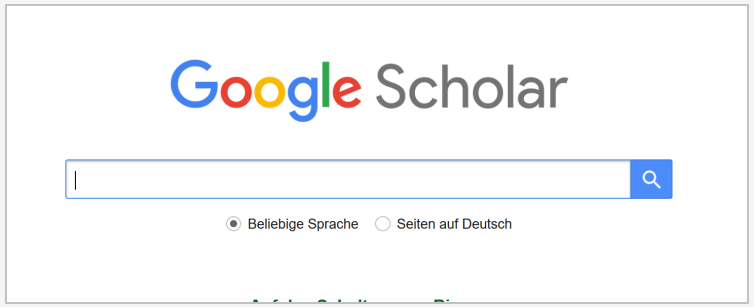
- **Verify** every claim in the primary source; cite the paper, never the assistant

Literature

How to Access Good Literature

STEP 1

Search on Google Scholar



STEP 2

Click "All 13 versions"

Below each result: publisher, preprint, author homepage and repository copies

STEP 3

Find the PDF

Pick a version with a [PDF] link

Behind a paywall? Use the **LMU library** e-media login and bookmarklet: it gives you licensed access from anywhere.

Literature

How to Read a Scientific Paper

A good paper

1. Abstract
2. Conclusion
3. Everything in between

A bad paper

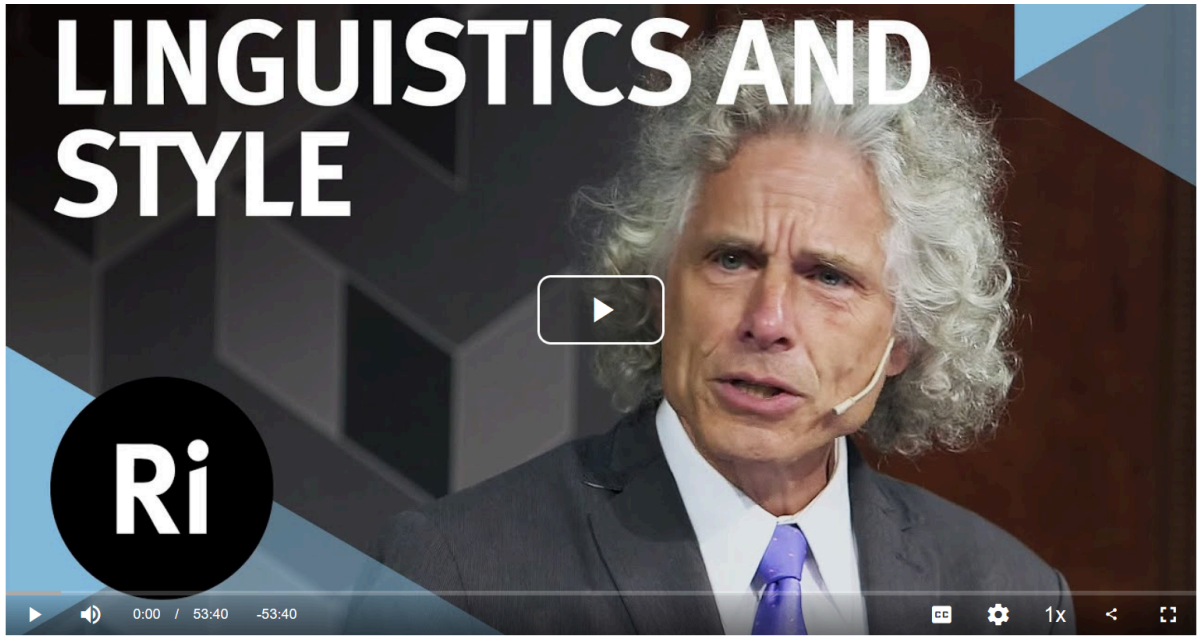
1. Abstract
2. Conclusion
3. *This is horrible. Why?*

> Bad papers teach you something too: **name what makes them bad**, and avoid it in your own paper.

Further reading: S. Keshav (2007), How to Read a Paper, ACM SIGCOMM Computer Communication Review 37(3).

Writing well is important!

- The paper is graded on **craft**: precise language, tidy graphics, clear wording and a pleasant reading flow
- Readers judge your science through your writing



Linguistics, Style and Writing in the 21st Century - with Steven Pinker 🎧

Watch: Steven Pinker, Linguistics, Style and Writing in the 21st Century, The Royal Institution.

Writing Well

Advice 1: Use Figures and Tables

amount	component	effect	measurement
1	diagram	saves	1000 words
1	table	replaces	3 paragraphs

Table 1: Effect of visual paper components on the amount of text

Table 1 shows the result of our text-reduction analysis: each component has a certain effect, measured in either paragraphs or words. We found that one diagram saves, on average, 1000 words ...

Every figure and table needs a **caption** and must be **referenced and explained** in the text.

~~Write and then write it again, better.~~

Write, then revise.

> If you aim for perfection in the first draft, you will not write anything. Get a draft down, then make the last draft **good enough**.

"The first draft of everything is shit."

attributed to Ernest Hemingway

"It is not about writing, it is about rewriting."

attributed to Anne Lamott

Writing Well

Advice 3: Don't Try to Sound Smart. Say Something.

TRYING TO SOUND SMART

If you try, by writing long-winded sentences, to enhance the text and make yourself appear more knowledgeable, the impression upon your reader will be such that none of the things you want to convey actually reaches their – however eagerly attentive – minds.

SAYING SOMETHING

Long-winded sentences lose the reader.

State your content directly, so that it reaches the reader and sticks.

Writing Well

Advice 4: With AI, Clear Writing Matters More

FLUENT TEXT NOW COSTS NOTHING. WHAT ONLY YOU CAN ADD:

1. **A clear point:** know what you want to say before you open a chatbot
2. **Taste:** write for one reader and one question; cut what does not serve them
3. **Responsibility:** stand behind every sentence, and disclose AI use (IEEE rule)

"Writing is thinking. We must teach students to construct arguments and analyze ideas, not simply offload the task to a machine."

Richard Socher, The Eureka Machine (2026)

An LLM picks the wording that fits the widest range of readers. A good writer chooses for one reader.

RUNNING EXAMPLE: TOPIC 14, LLM ROUTING

Generated: In today's rapidly evolving AI landscape, LLM routing is not just an optimization; it's a paradigm shift. This paper delves into the intricate interplay of cost, quality and latency.

Clear: A router sends each query to the cheapest model that can answer it. This paper measures how much cost three routers save and how much answer quality they lose.

THEN EDIT OUT THE SIGNS OF AI WRITING

Not X but Y	Empty significance	Staged run-ups	AI words
"This is not just a benchmark; it is a new paradigm."	*"At its core, routing is about trust."* *"The implications are profound."*	*"Here's the thing: cost matters."*	*"This study delves into the landscape and underscores …"*

Readers notice: at least 13.5% of 2024 PubMed abstracts were processed with an LLM, and "delves" appeared 28 times more often than expected.

R. Socher, The Eureka Machine, PublicAffairs (2026); Wikipedia, Signs of AI writing; blader/humanizer (GitHub); Kobak et al., Science Advances 11 (2025); IEEE, Guidelines for AI-generated content.

Wrap-up

Next Deadline: First Version of Your Paper

PAPER SKELETON

0 Abstract
1 Introduction
2 Background and Related Work
3 Your topic-specific sections ...
4 ... agreed with your mentor
5 Conclusion
Bibliography

Paper due
The date is announced on Moodle

Use the **IEEE LaTeX template** from the start: 6 to 8 pages.